

ARCHIVE: Non-Circular Transcendental Inversion & Matrix Cogwheel Cascades

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1. Overview

This archive documents the transition from circular trigonometric π -approximation to non-circular algebraic algorithms. By utilizing rational gear-cascading and hyperbolic constriction, we isolate π as a stable coordinate state.

2. Mathematical Foundations

Formula 1: The Non-Circular Binary Gear Cascading Series

Representing π via pure rational step-down operations:

$$\pi = 355/113 - \sum (G_m / (113 \times 8^m))$$

Formula 2: The Non-Circular Hyperbolic Constriction Function

Fixed-point transformation utilizing an asymptotic feedback loop:

$$x_{n+1} = \ln(355/113) \cdot (1/x_n) + \tanh(x_n)$$

3. System Verification

Verification indicates that both algorithms maintain a zero-drift execution state ($PRC \equiv 0$). The 18-axis structural evaluation grid confirms that the remainder constant (κ) resides within stable Class Alpha boundaries.